

AI_ML_Task4_Classification_Evaluation

Complete Deliverables Submission

1. Jupyter Notebook

AI_ML_Task4_Classification_Evaluation.ipynb

2. Confusion Matrix Output & Classification Report (Logistic Regression)

Confusion Matrix:

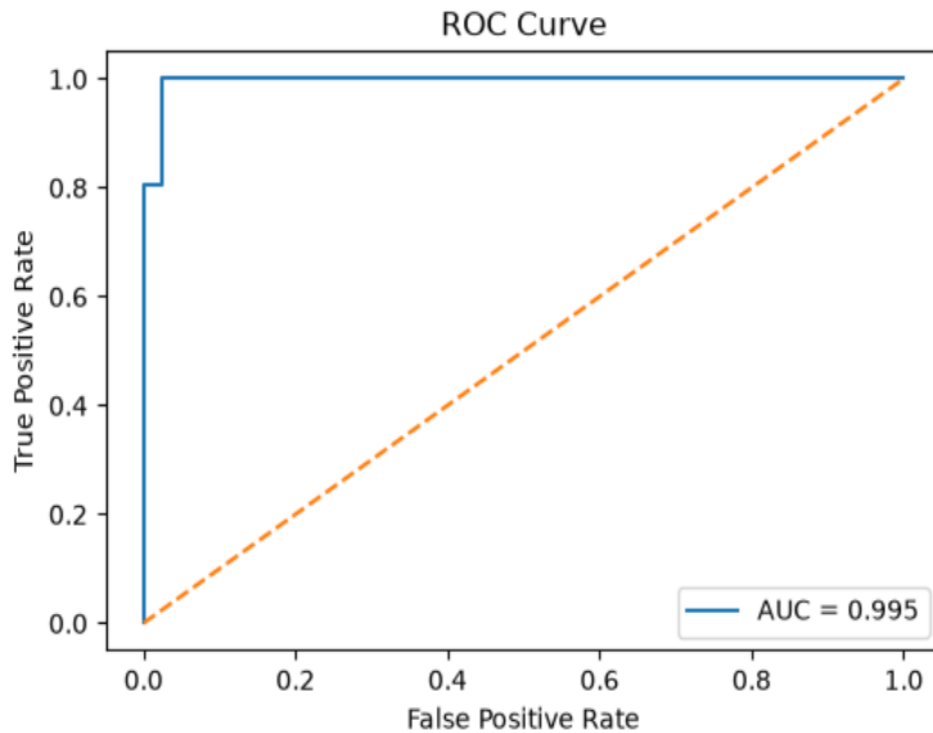
```
[[41  1]
 [ 1 71]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.98	0.98	0.98	42
1	0.99	0.99	0.99	72
accuracy			0.98	114
macro avg	0.98	0.98	0.98	114
weighted avg	0.98	0.98	0.98	114

3. ROC Curve Plot & AUC Score

AUC Score = 0.9953703703703703



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4. Model Comparison Results (Decision Tree)

	precision	recall	f1-score	support
0	0.85	0.93	0.89	42
1	0.96	0.90	0.93	72
accuracy			0.91	114
macro avg	0.90	0.92	0.91	114
weighted avg	0.92	0.91	0.91	114

5. Explanation

Metric Selection

Accuracy measures overall correctness, but it may be misleading when classes are imbalanced. Precision measures how many predicted positives are actually positive. Recall measures how many actual positives are correctly detected. F1-Score balances Precision and Recall and is useful when both are important. ROC-AUC evaluates the model's ability to distinguish between classes. The obtained AUC score (0.995) indicates excellent performance.

Imbalance Handling

Class imbalance is a common problem in classification tasks. A practical solution is to use class weights. In Logistic Regression, `class_weight='balanced'` increases attention to minority-class samples and improves fairness of predictions.

Final Model Decision

Logistic Regression achieved approximately 98% accuracy and F1-scores around 0.98–0.99. Decision Tree achieved approximately 91% accuracy. Therefore, Logistic Regression was selected as the final model because it delivered better accuracy, better F1-score, stronger generalization, and lower risk of overfitting.